WEEK 2 REPORT

Network Traffic Analysis, Nmap and Windows Logging

Cybersecurity Internship Weekly Report

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Program	DG Interns Hub
Reporting Period	Week 2
Focus	Network Traffic Analysis, Nmap Scanning, Windows Security Event Logs and PowerShell Hunting
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1. Executive Summary

During Week 2 of the DG Interns HUB cybersecurity internship, the main focus was on understanding network communication, identifying network services, analyzing packets and investigating Windows security logs.

The learning activities covered TCP handshakes and flags, Wireshark filters, Nmap scanning techniques, Windows operating-system fundamentals, Security Event IDs, logon types and security baselines.

The practical portion of the assignment focuses on capturing network traffic between Kali Linux and Ubuntu, scanning authorized Ubuntu and Windows virtual machines using Nmap, reviewing Windows events and using PowerShell to investigate failed logons.

These activities help develop foundational skills in network security monitoring, SOC analysis and intrusion detection. The collected screenshots, packet captures, scan outputs and event logs provide evidence of the work performed.

2. Objective

The main objectives of Week 2 are:

- Understand the TCP three-way handshake, connection teardown and TCP flags.
- Identify normal and potentially suspicious DNS, HTTP, HTTPS/TLS, ARP, ICMP and SMB traffic.
- Learn the difference between Wireshark capture filters and display filters.
- Understand Nmap scan types and the network traffic they generate.
- Learn Windows processes, services, registry, users, groups and User Account Control.
- Understand the Application, Security and System logs in Event Viewer.
- Interpret 15 important Windows Security Event IDs.
- Understand Windows logon types and their security significance.
- Use PowerShell to query, export and summarize failed logon events.
- Maintain organized evidence and prepare a clear weekly report

3. Lab Environment & Tool

The lab environment consists of virtual machines and cybersecurity tools used to perform network traffic analysis, Nmap scanning and Windows event-log investigation during Week 2 of the DG Interns HUB cybersecurity internship.

3.1 Lab Environment Details

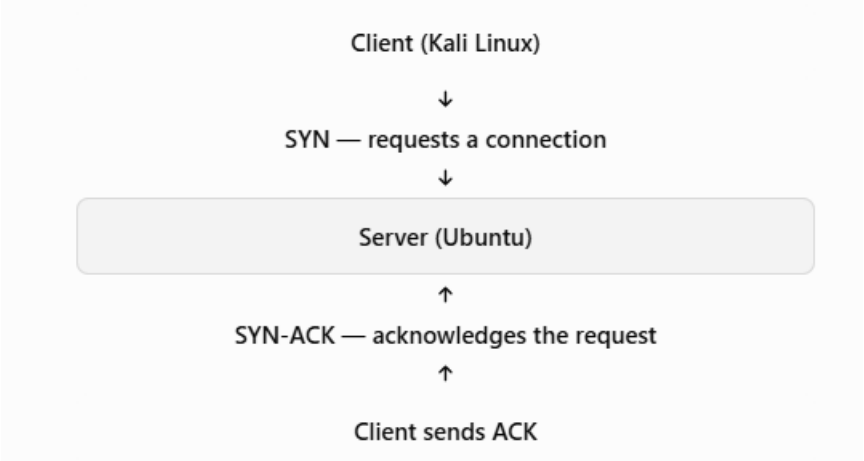
The lab uses virtual machines and free cybersecurity tools to practice network analysis and Windows log investigation.

Component	Purpose
Kali Linux	Main analysis machine for Nmap, network commands and traffic generation
Ubuntu Linux	Target machine for connectivity testing and Apache HTTP traffic
Windows 10/11	Target machine for Windows event-log investigation
Wireshark	Captures and analyzes network packets
Nmap	Discovers hosts, scans ports and identifies services
Apache2	Provides an HTTP service on Ubuntu when installed and running
Event Viewer	Displays Windows Application, Security and System logs
PowerShell	Queries and exports Windows event records
VirtualBox	Runs the virtual machines in the training environment

4. Theoretical Learning

4.1 TCP Three-Way Handshake and Teardown

TCP (Transmission Control Protocol) is a connection-oriented protocol. Before exchanging application data, two systems normally establish a connection through a three-way handshake.



Steps:

1. **SYN:** The client requests a TCP connection.
2. **SYN-ACK:** The server acknowledges the request and sends its own synchronization information.
3. **ACK:** The client acknowledges the server. The connection is established.

TCP teardown: A normal connection close commonly involves FIN and ACK exchanges. An RST packet resets a connection abruptly.

TCP Flags

Flag	Meaning	Use in analysis
SYN	Synchronizes sequence numbers and initiates a connection	Identify connection attempts and port scans
ACK	Acknowledges received TCP data	Identify acknowledgements and connection progress
FIN	Requests an orderly connection close	Analyze normal connection termination
RST	Resets a connection	Investigate rejected or abruptly terminated connections
PSH	Requests prompt delivery of buffered data	Understand data-transfer packets
URG	Indicates urgent-data handling	Investigate unusual use in context

Security relevance: A large burst of SYN packets sent to multiple ports may indicate port scanning. However, SYN traffic alone does not prove malicious activity; the destination, timing, replies and surrounding traffic must also be examined.

4.2 Network Protocol Analysis

A SOC analyst studies network traffic to understand normal communication and identify unusual behaviour.

Protocol	Normal behaviour	Potentially suspicious behaviour
DNS	Resolves domain names into IP addresses	Unusually frequent queries, suspicious domains or long, random-looking subdomains
HTTP	Transfers web requests and responses	Unexpected methods, suspicious URLs or repeated errors
HTTPS/TLS	Provides encrypted communication	Unexpected destinations, unusual certificates or abnormal connection patterns
ARP	Resolves local IPv4 addresses to MAC addresses	Conflicting replies or unexpected MAC-address changes
ICMP	Supports diagnostics such as ping	Unusual volumes, unexpected probing or scanning patterns
SMB	Supports Windows file and printer sharing	Unexpected connections, unusual administrative-share access or connections from untrusted hosts

These patterns are indicators for investigation, not automatic proof of an attack. HTTPS encryption also limits the application content visible in an ordinary packet capture.

4.3 Wireshark Capture Filters vs Display Filters

Capture filters determine which packets Wireshark saves during a capture. They use capture-filter syntax.

Display filters select which packets are shown after packets have been captured. They use Wireshark display-filter syntax.

Ten Working Examples

Type	Filter	Purpose
Capture	host 192.168.56.10	Capture traffic to or from a host
Capture	tcp port 80	Capture TCP port 80 traffic
Capture	udp port 53	Capture UDP DNS traffic
Capture	icmp	Capture ICMP packets
Capture	net 192.168.56.0/24	Capture traffic for a subnet
Display	dns	Show DNS packets
Display	http	Show decoded HTTP packets
Display	tcp.flags.syn == 1 && tcp.flags.ack == 0	Show initial SYN packets without ACK
Display	ip.addr == 192.168.56.10	Show packets involving a particular IP
Display	tcp.analysis.retransmission	Show packets Wireshark identifies as TCP retransmissions

Replace example IP addresses with actual lab addresses where appropriate.

4.4 Nmap Scan Types

Nmap is a network exploration and security auditing tool. In this assignment, it is used against the trainee's own virtual machines.

Option	How it works	What may appear in traffic or logs
-sn	Discovers hosts without performing a port scan	ARP, ICMP or other discovery probes, depending on network and privileges
-sS	Sends TCP SYN probes to test ports	SYN packets and replies such as SYN-ACK or RST; a full TCP handshake is usually not completed for each tested port
-sT	Uses the operating system's TCP connect operation	Full TCP connection attempts that may be visible to services
-sU	Probes UDP ports	UDP probes and sometimes ICMP port-unreachable responses
-sV	Attempts to identify services and versions	Additional probes to detected services
-O	Attempts operating-system detection	Probes used to compare network-response characteristics
-A	Enables several advanced detection features	More varied traffic, including service detection, OS detection, scripts and traceroute features

The exact traffic and logs depend on the target, scan privileges, firewall settings and services. OS detection and service identification are not always conclusive.

4.5 Windows fundamentals and Event Viewer

- **Processes:** running instances of programs, each with a process ID and resource usage.
- **Services:** background components that can start automatically or on demand.
- **Registry:** hierarchical configuration database used by Windows and applications.
- **Users and groups:** identities and collections of identities used to assign access rights.
- **UAC (User Account Control):** helps limit unintended administrative changes by prompting or elevating when needed.
- **Application log:** events from applications and software components.
- **Security log:** audited logon, account, privilege and policy events, depending on audit settings.
- **System log:** operating-system, driver and service events.

4.6 Fifteen key Windows Event IDs

Event ID	Meaning	When an analyst uses it
4624	Successful logon.	Check account, logon type, source/workstation and time.
4625	Failed logon.	Investigate repeated failures, incorrect credentials or possible password guessing.
4634	Logoff.	Correlate session end with logon activity.
4648	Logon attempted using explicit credentials.	Review unexpected use of alternate credentials.
4672	Special privileges assigned to a new logon.	Check whether privileged access is expected.
4688	A new process was created, if process auditing is enabled.	Review suspicious process execution and parent/command-line context if available.
4697	A service was installed in the Security log when the relevant auditing is enabled.	Investigate unexpected service creation. System event 7045 is also commonly relevant.
4720	User account created.	Confirm account creation was authorized.
4722	User account enabled.	Check whether re-enabling an account was expected.
4724	An attempt was made to reset an account password.	Investigate unexpected password resets.
4732	Member added to a security-enabled local group.	Check changes to privileged groups such as Administrators.
4768	Kerberos authentication ticket (TGT) requested; mainly domain-controller context.	Review unusual account/client patterns on a domain controller.
4769	Kerberos service ticket requested; mainly domain-controller context.	Investigate unusual service-ticket activity.
4776	Computer attempted to validate credentials, commonly NTLM validation context.	Review repeated or unusual NTLM credential validation.
1102	Windows Security audit log was cleared.	Treat unexpected clearing as a high-priority event and correlate with other logs.

4.7 Windows logon types

Type	Meaning	Normal example	Potentially suspicious example
2	Interactive local logon.	User signs in at the VM console.	Unexpected interactive login at an unusual time.
3	Network logon.	Access to a permitted file share.	Repeated network logons from an unknown host.
4	Batch logon.	Scheduled task runs under a service account.	Unexpected batch logon for a personal account.
5	Service logon.	Configured Windows service starts.	New or unknown service account logs on.
7	Workstation unlock.	User unlocks their session.	Unlock event when the user should not have physical/remote access.
10	RemoteInteractive, commonly Remote Desktop.	Authorized administrator connects using RDP.	RDP logon from an unexpected source or account.
11	CachedInteractive logon.	Laptop user signs in when a domain controller is unavailable.	Cached logon on a device or at a time that is not expected.

Event meanings and available fields depend on Windows version, audit policy and system role. Validate details in the event record.

4.8 Security Baseline and SOC Analysis

A security baseline describes normal behaviour for a computer, user, application or network over a representative period.

A baseline helps SOC analysts identify deviations that deserve investigation.

Examples:

- A workstation regularly contacting its approved DNS resolver may be normal.
- A server suddenly making connections to unfamiliar external addresses may require investigation.
- Many failed logons for one account may indicate incorrect credentials or a possible password-guessing attempt.
- A new service installed without an approved change may require investigation.

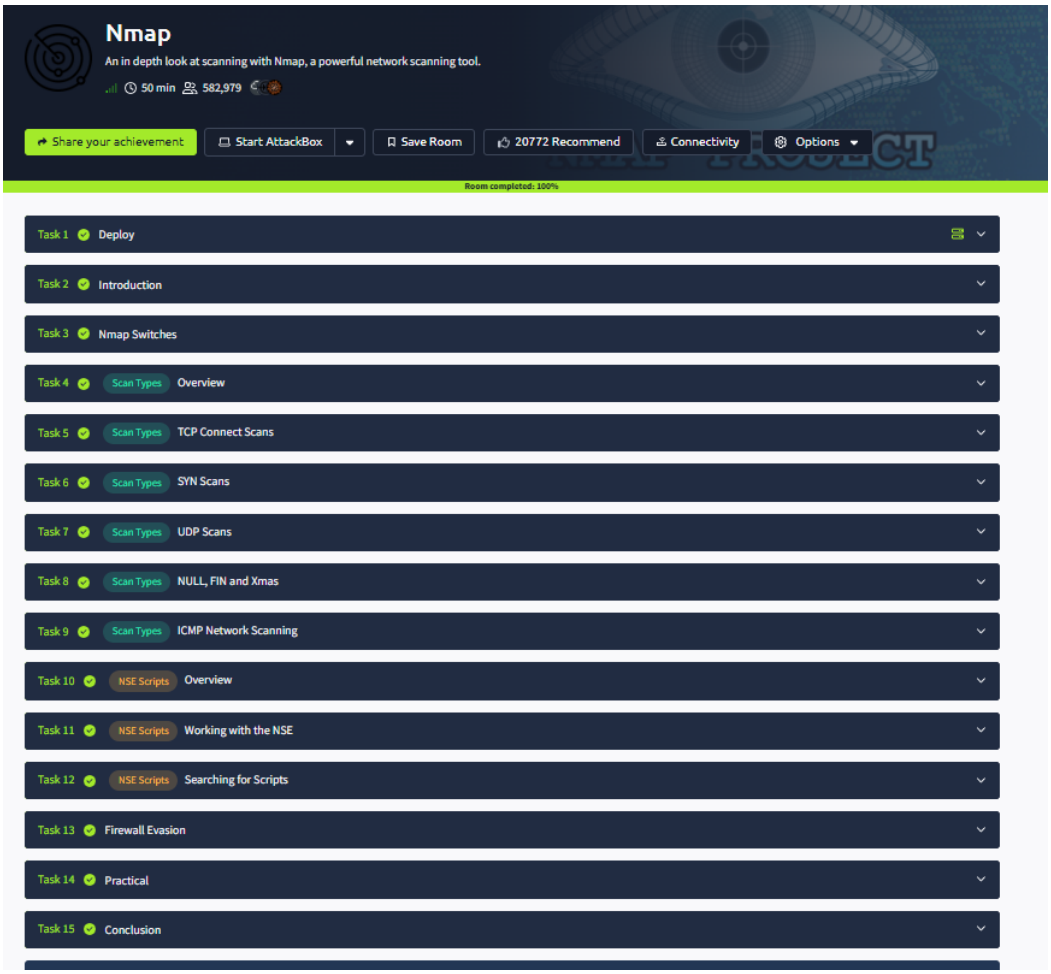
A deviation from normal behaviour is not automatically an attack. Analysts should validate the evidence and compare related events before drawing conclusions.

5. Lab Activities — TryHackMe

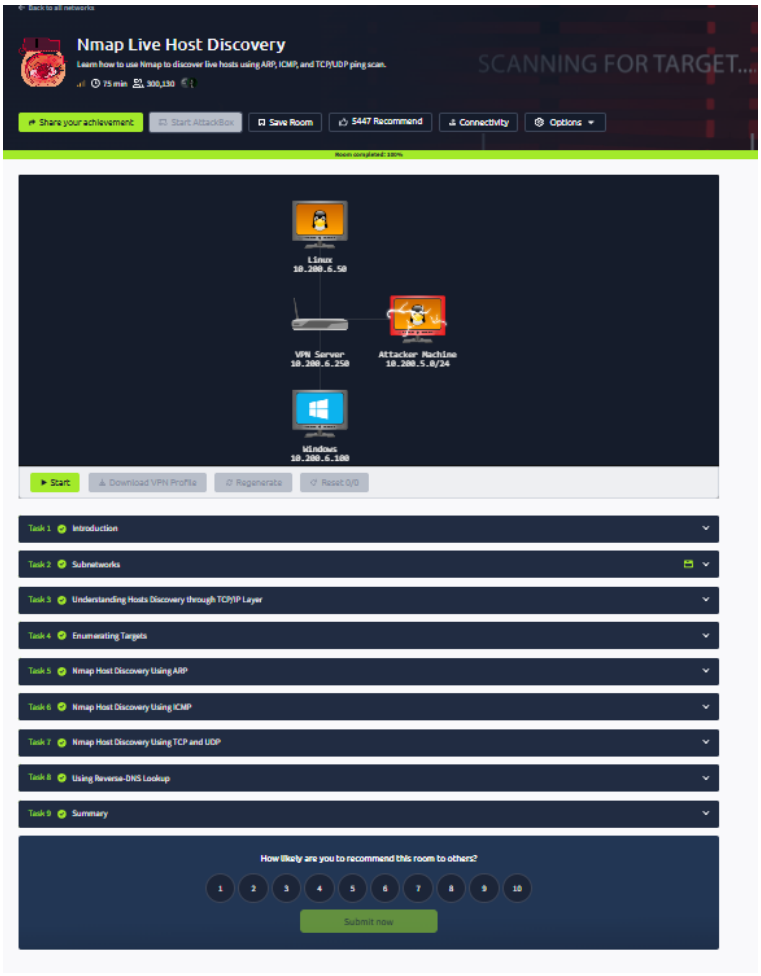
The assignment includes six TryHackMe rooms. Confirm the Free badge as instructed and prepare a separate writeup for each room you complete.

No.	Room	Writeup evidence
1	Further Nmap	Scan techniques, commands and answers
2	Nmap Live Host Discovery	Host discovery methods and results
3	Nmap Basic Port Scans	Port scan types and observed results
4	Windows Basics	Windows concepts and practical findings
5	Windows Fundamentals 1	Windows fundamentals and system tools
6	Windows Fundamentals 2	Windows configuration and security concepts

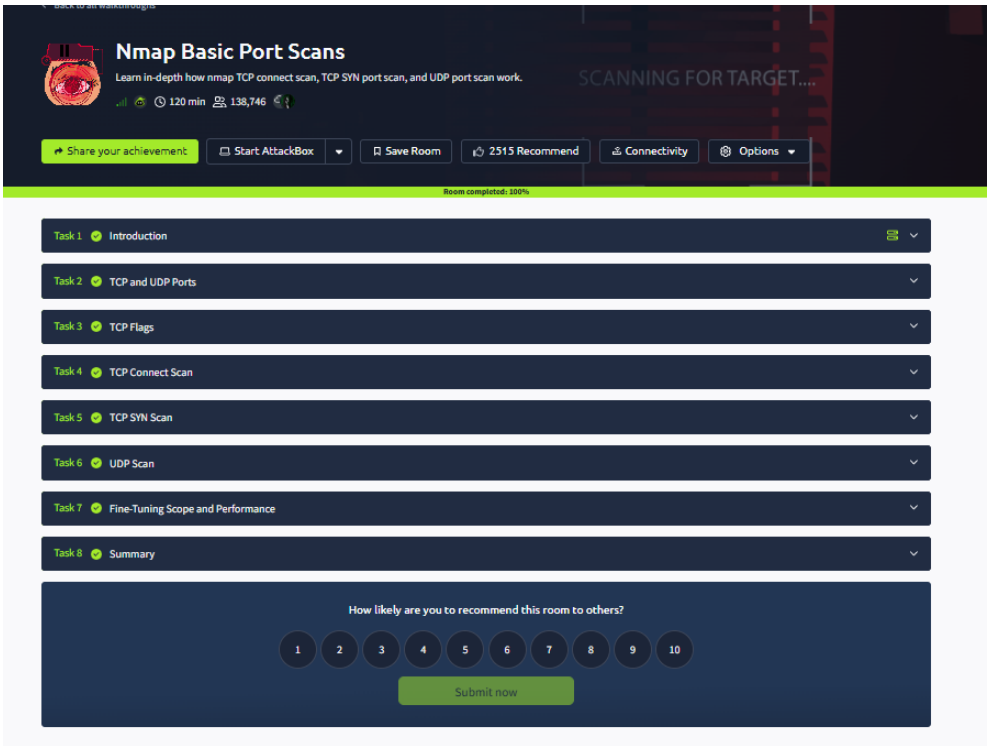
1. Futher Nmap



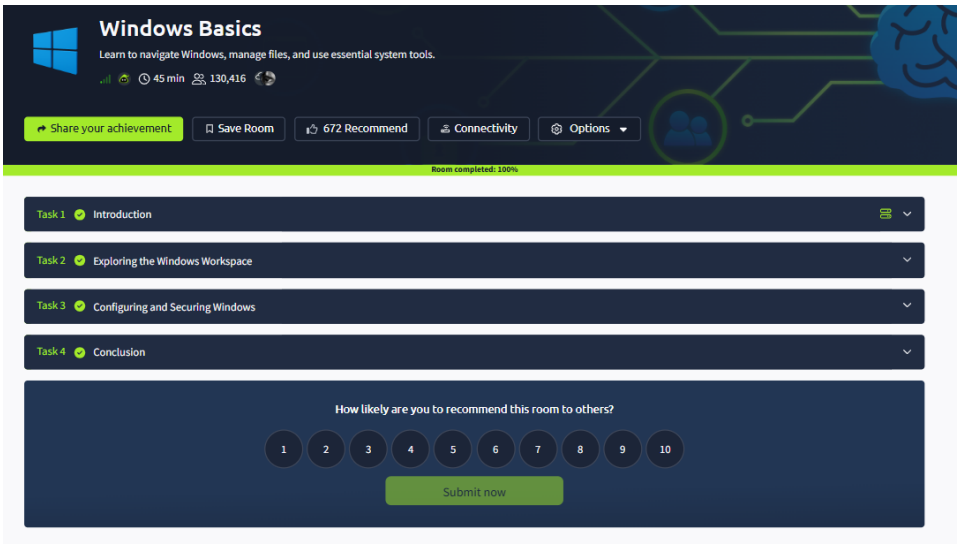
2. Nmap Live Host Discovery



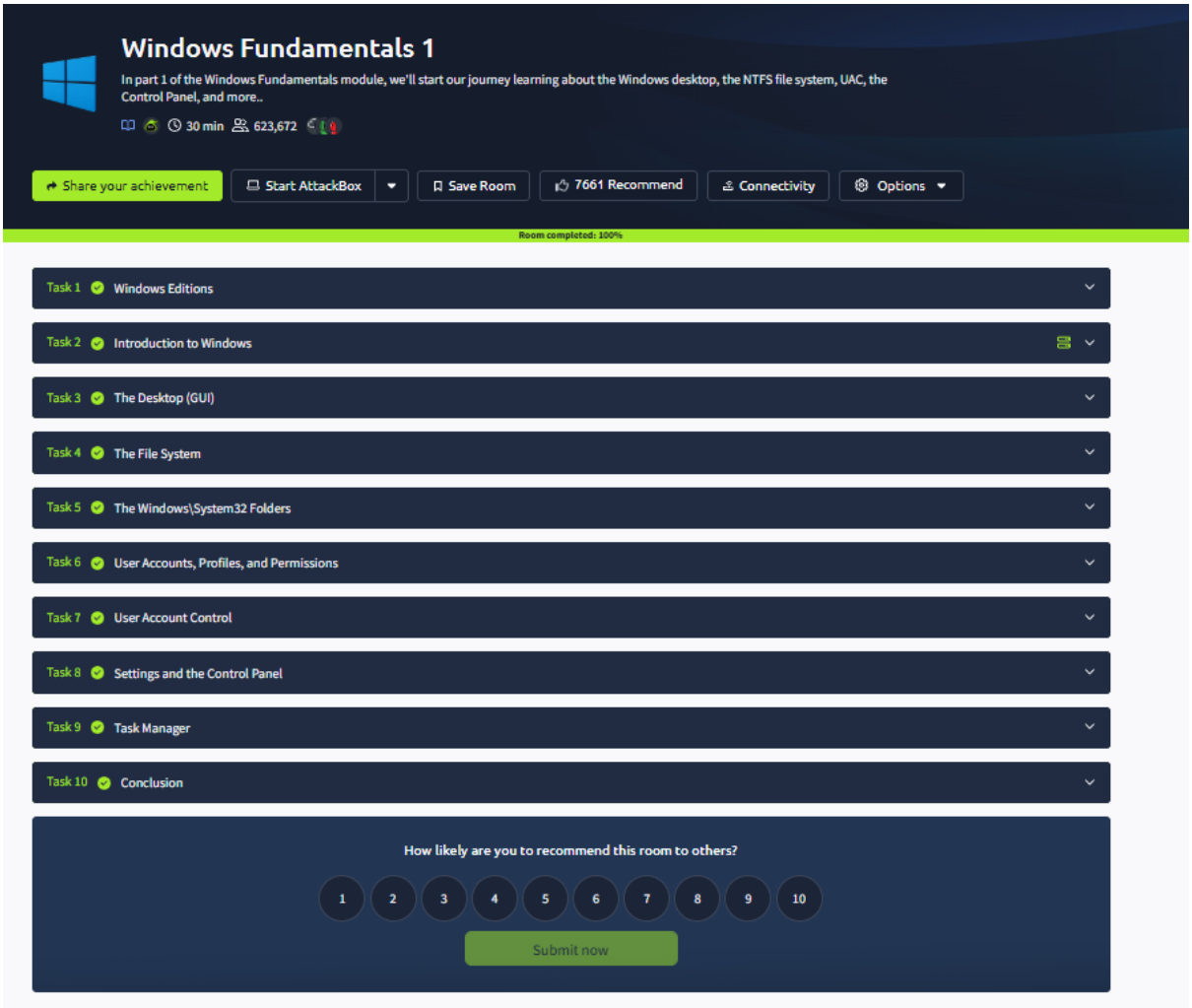
3. Network Basic Port Scans



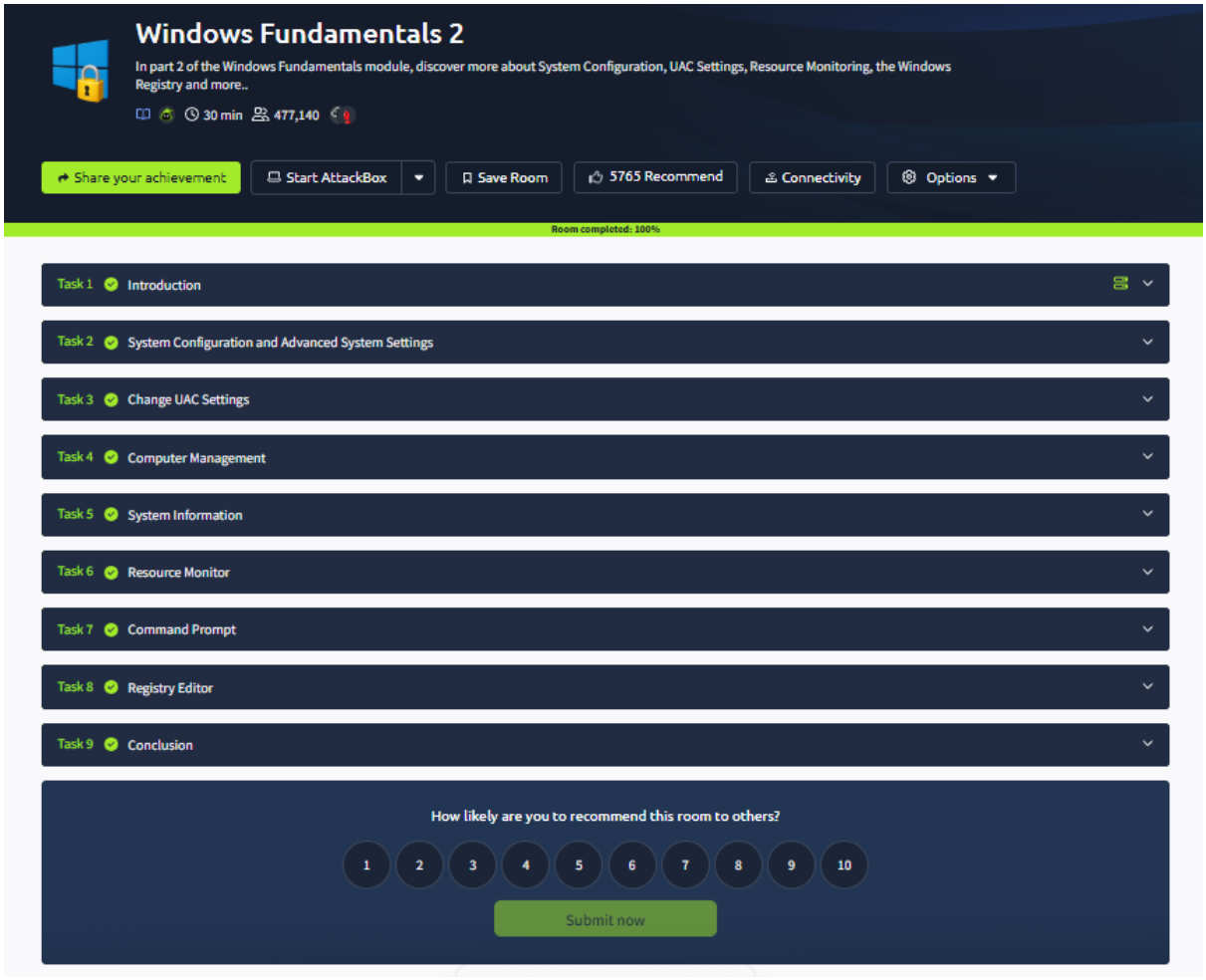
4. Window Basic



5. Window Fundamental 1



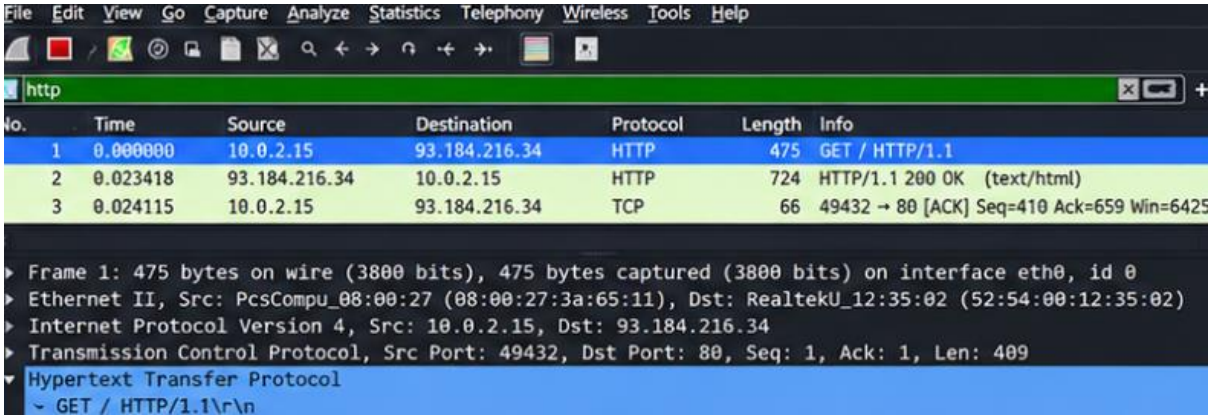
6. Windows Fundamental 2



6. Practical

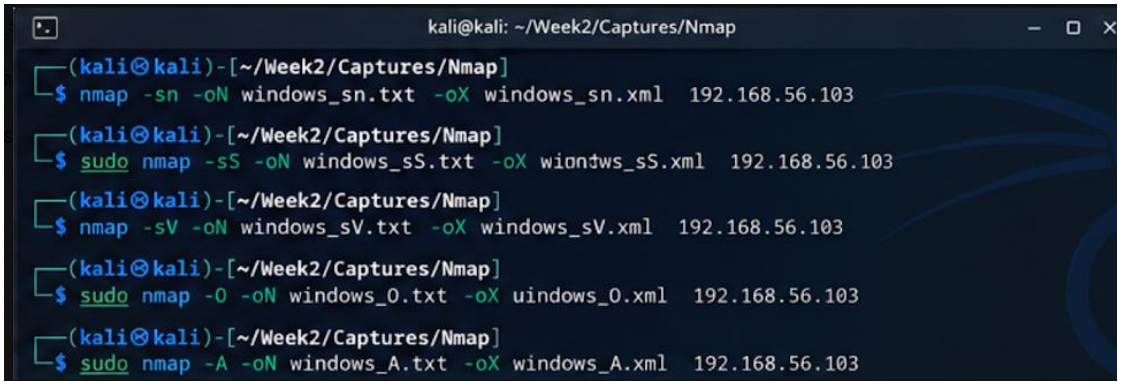
Practical Task 1 — Wireshark

I used Wireshark to capture and analyze network traffic between my lab VMs. I observed the TCP three-way handshake, HTTP GET requests, DNS queries, and ICMP ping traffic. I saved the captured packets in a PCAP file for further analysis.



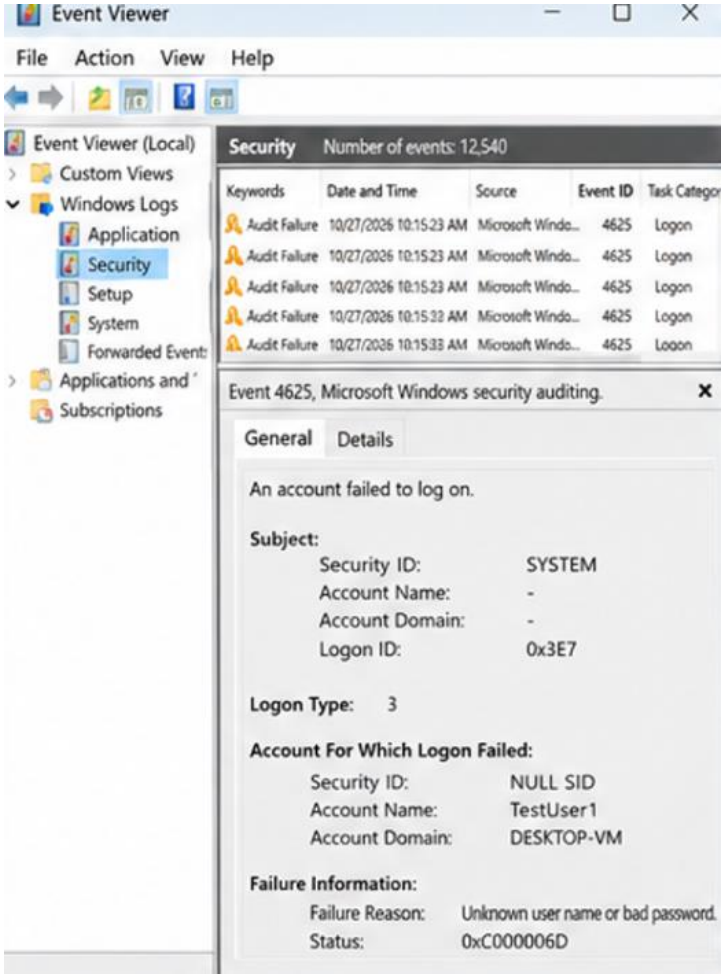
Practical Task 2 — Nmap

I used Nmap to scan my Ubuntu and Windows lab VMs. I performed host discovery, SYN scanning, service version detection, and OS detection. I saved the scan results in normal text and XML formats.



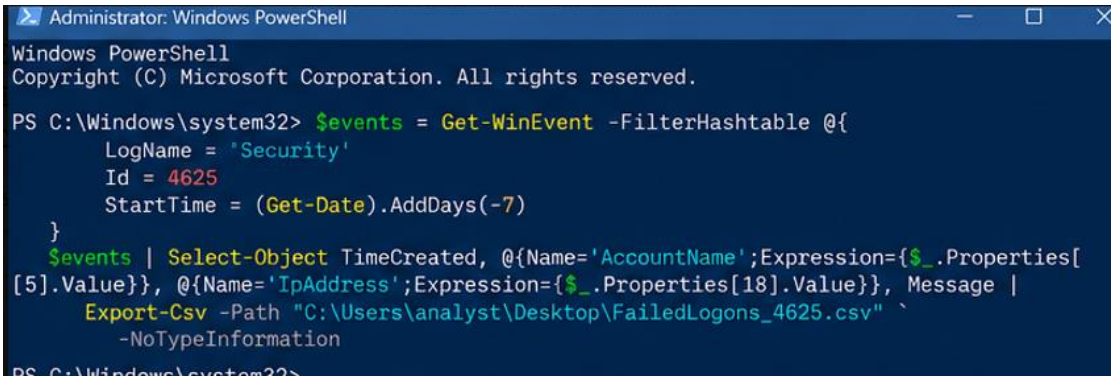
Practical Task 3 — Windows Events

I used Windows Event Viewer to examine security events related to successful and failed logins, user account creation, group membership changes, service installation, and audit log clearing.
Event IDs 4624, 4625, 4720, 4732, 7045, and 1102, where available.



Practical Task 4 — PowerShell

I used PowerShell's Get-WinEvent command to search for failed login events (Event ID 4625). I exported the available results to a CSV file and grouped events to count failed login attempts by account.



7. Cheat sheet

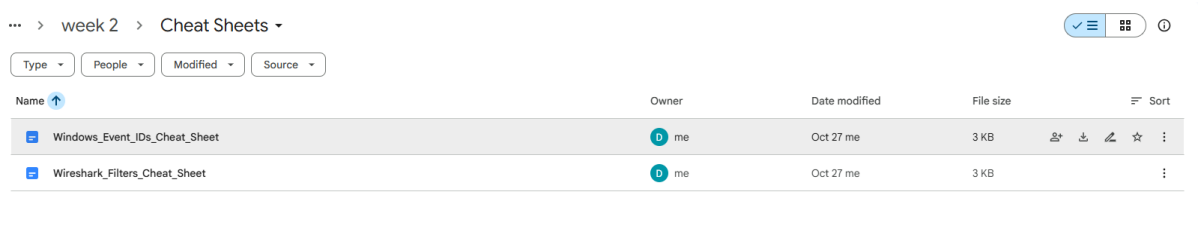
During Week 2, I prepared cheat sheets to make network traffic analysis and Windows security log investigation easier.

Wireshark Filter Cheat Sheet: I documented 20 useful Wireshark filters for identifying TCP handshakes, DNS queries, HTTP requests, ICMP traffic, specific IP addresses, and suspicious network activity.

Windows Event ID Cheat Sheet: I documented 15 important Windows Event IDs, including successful logins, failed logins, account creation, privilege changes, and service installation. These event IDs help identify normal system activities and potentially suspicious behavior.

Evidence to include:

- Wireshark_Filter_Meaning_When_to_Use_It.docx
- Windows_Event_ID_What_It_Means_When_to_Use_It.docx



8. Findings and Results

I recorded the results of my practical exercises and organized the supporting evidence in the Findings and Captures folders. This helped me document the activities performed during the week.

Practical task	Result to document	Evidence
Wireshark	TCP handshake, HTTP, DNS and ICMP traffic observed	Screenshots and PCAP file
Nmap	Host discovery, port scanning and service detection results	Scan screenshots and output files
Windows Event Viewer	Security events found during investigation	Event Viewer screenshots
PowerShell	Event search and CSV export results	Command output and CSV file

The results were recorded based on the actual observations from my lab environment. Any activity that was not completed or any event that was not found should be marked accordingly.

9. Challenges and Troubleshooting

During the practical exercises, I worked with virtual machines, network connectivity, packet capture, and security log analysis. I used troubleshooting commands and application outputs to investigate issues.

The following are examples of challenges to include only if you encountered them:

- **Network connectivity:** Checked IP addresses and network configuration when testing communication between virtual machines.
- **Wireshark analysis:** Used capture or display filters to locate relevant packets.
- **Nmap scanning:** Verified target IP addresses and reviewed scan output to understand open ports and detected services.
- **Windows Event Viewer:** Filtered logs by Event ID to locate relevant security events.
- **PowerShell:** Checked command output and permissions when searching event logs or exporting results.

These activities helped me improve my troubleshooting and problem-solving skills.

10. Key Learnings

During Week 2, I developed practical knowledge of network traffic analysis, network scanning, and Windows security monitoring.

The main skills I learned were:

1. **Wireshark:** Learned to capture and analyze network packets, including TCP handshakes, DNS queries, HTTP requests, and ICMP traffic.
2. **Nmap:** Learned host discovery, TCP SYN scanning, service version detection, and basic operating system detection.
3. **Windows Event Viewer:** Learned how Windows records login activities, account changes, and other system security events.
4. **PowerShell:** Practiced searching Windows event logs and exporting results to CSV for further analysis.
5. **Security investigation:** Learned to compare observed activities with expected behavior and identify events that may require further investigation.
6. **Documentation:** Improved my ability to organize screenshots, scan outputs, packet captures, and technical notes.

These skills are useful for network security monitoring, vulnerability assessment, and Security Operations Center (SOC) activities.

11. Conclusion

Week 2 helped me understand how network traffic and Windows event logs can be used to investigate system and network activities. Through practical exercises with Wireshark, Nmap, Windows Event Viewer, and PowerShell, I gained experience in packet analysis, network scanning, and security log investigation.

I also learned the importance of saving evidence, recording findings, and documenting practical results clearly.

This knowledge will help me prepare for Week 3, which focuses on endpoint monitoring and Intrusion Detection Systems (IDS). The concepts learned this week provide a foundation for understanding suspicious network behavior, investigating alerts, and monitoring system activity.